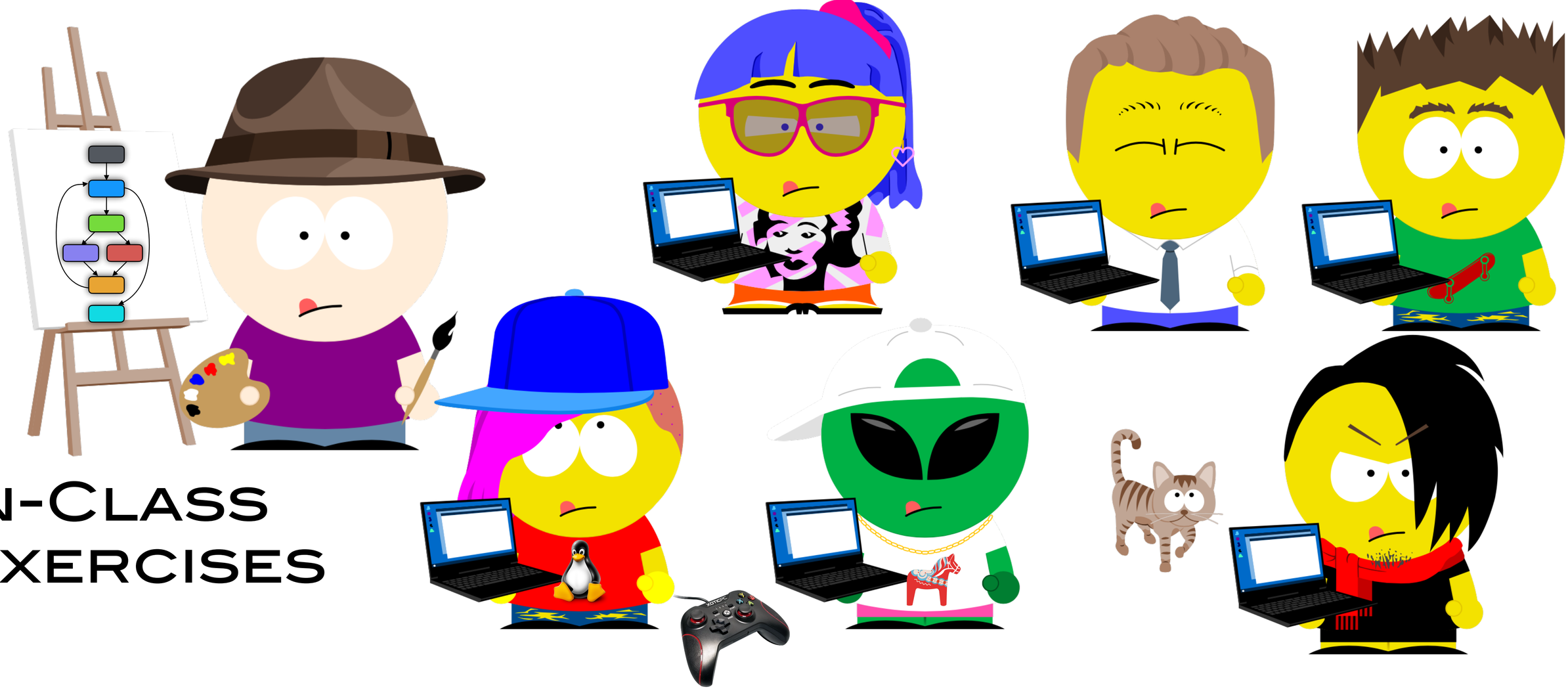


In-Class Exercises

Version: 2026-04-06

Intermediate Code 3 –Control Flow Graphs

Download exercises

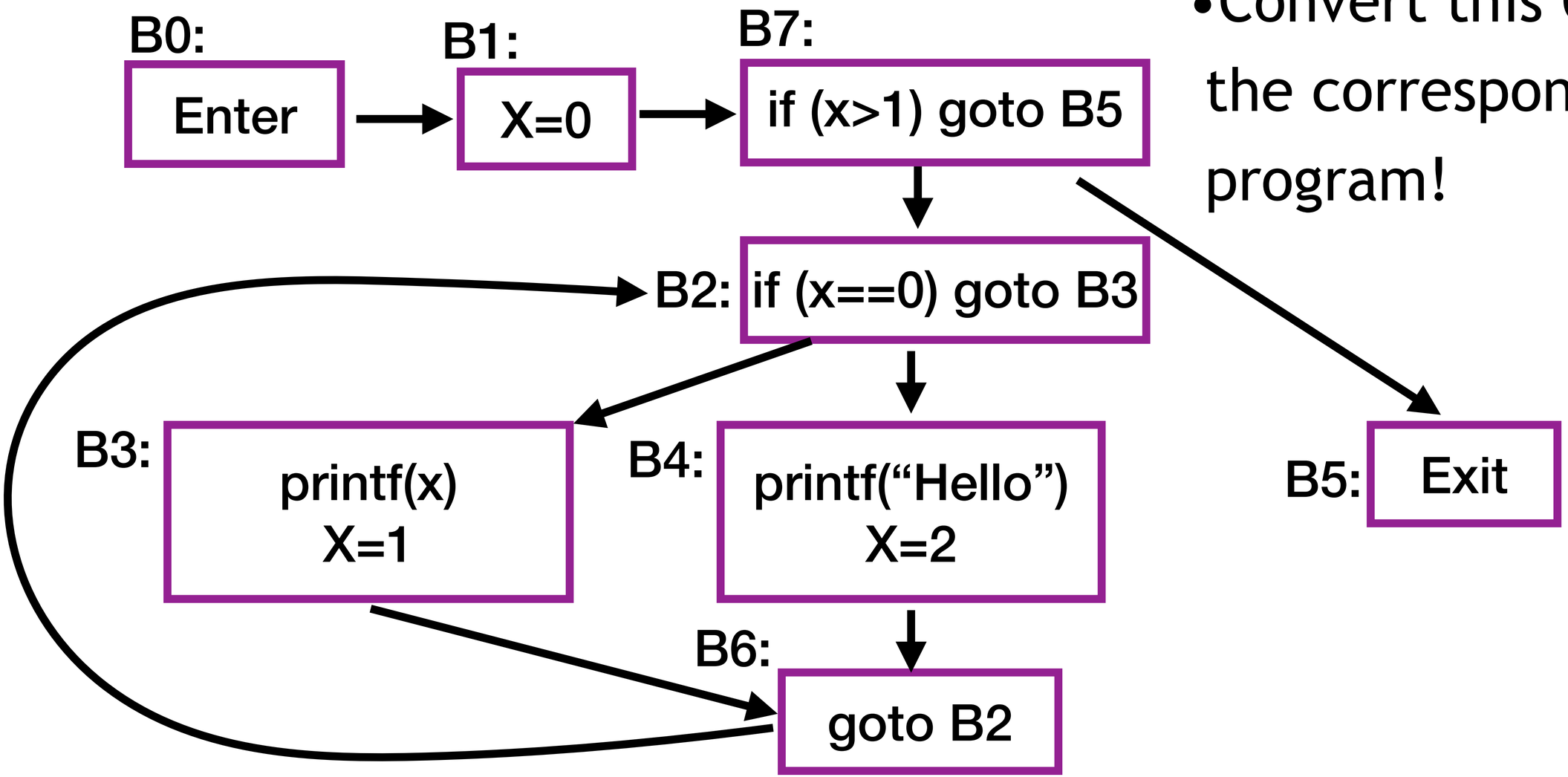
1. Inside the LigerLabs VM, open up the Firefox browser and go to
   https://ligerlabs.org/compilers.html
2. Download the file
   `interm-3-exercises.zip`
3. Open up a terminal and Unzip the file

```
> unzip interm-3-exercises.zip
> cd interm-3-exercises
> ls
```

Recognizing CFGs

Task 1: Decompile

- Convert this CFG to the corresponding C program!



Task 2: Construct CFG

- Given this sequence of 3-address statements, mark the leaders and identify the basic blocks!

```
(1)  X=20
(2)  if X >= 10 goto (8)
(3)  X=X-1
(4)  A[X] = 10
(5)  if X != 4 goto (7)
(6)  X=X-2
(7)  goto (2)
(8)  Y=X+5
```

Task 3: Construct CFG

- Following the steps of the CFG-construction algorithm, convert this function to a CFG.

```
int gcd(int x, int y) {
    int temp;
    while (true) {
       if (x%y == 0) break;
        temp = x%y;
        x = y;
        y = temp;
    }
}
```

Task 4: Pair Challenge

- Sketch a small C program with very complex control flow.
- Transform the program to a CFG.
- Switch CFGs with the partner to the left of you!
- Who can transform the program they've been given back to C the fastest?

Task 5: Exceptions

- Consider the Java program on the next slide.
- It will throw a `java.lang.ArithmeticException` exception when a division by zero occurs.
- What would a CFG for the `main` function look like?

```
class Exc {
void main (String[] args) {
    int c = args.length;
    try {
      int i = c;
      while (i <= c) {
         int x = 42/i;
         i--;
      };
    } catch (java.lang.ArithmeticException e) {
      System.out.println("DIV-BY-ZERO");
    };
}
}
```

Definitions and Algorithms

2. Next, execute this algorithm to break the sequence of instructions into basic blocks (the nodes of the control-flow graph) and graph edges between these blocks:

a. Mark every instruction which can start a basic block as a leader:

- the first instruction is a leader;
- any target of a branch is a leader;
- the instruction following a conditional branch is a leader.

b. A basic block consists of the instructions from a leader up to, but not including, the next leader.

c. Add an edge A → B if A ends with a branch to B or can fall through to B.

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